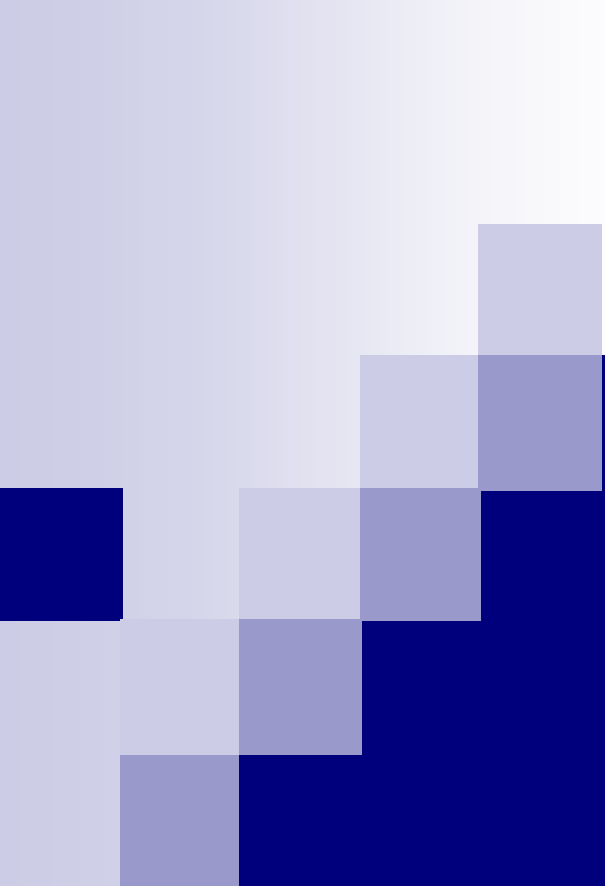


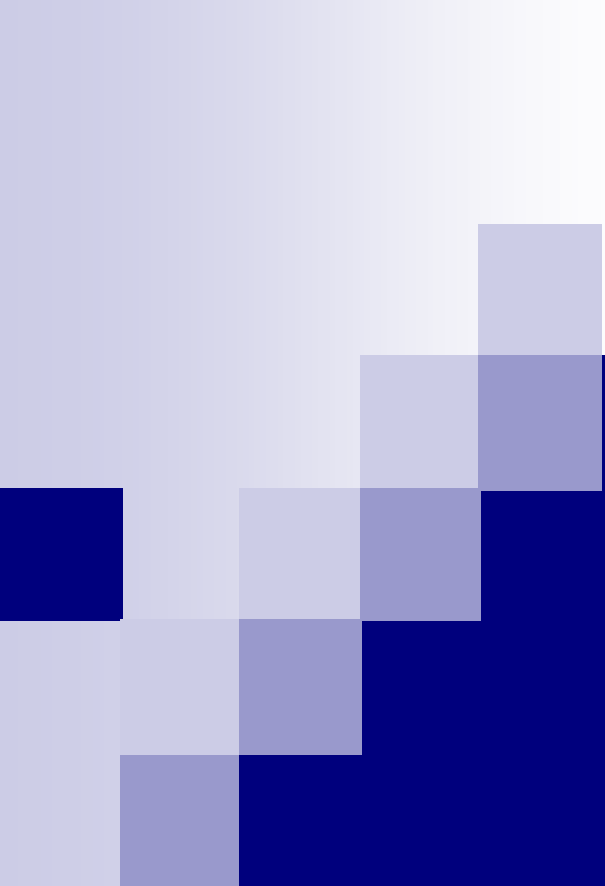


Process Assurance

Lecture # 38

Today's Lecture

- In our discussions on software quality assurance, we have divided the quality assurance into two major aspects
 - Process-related
 - Product-related
- Today, we're going to talk about process-related issues of software quality and discuss process assurance



Process Assurance



PUACP

Process Assurance - 0

- Process assurance makes certain that the process for building and delivering software is robust and allows for the delivery and maintenance of the products

Process Assurance - 1

- Process assurance consists of the collective activities carried out while developing a product to ensure that the methods and techniques used are integrated, consistent, and correctly applied
- Emphasis is given to cost, time, technical requirements, testing measurements, and prototyping

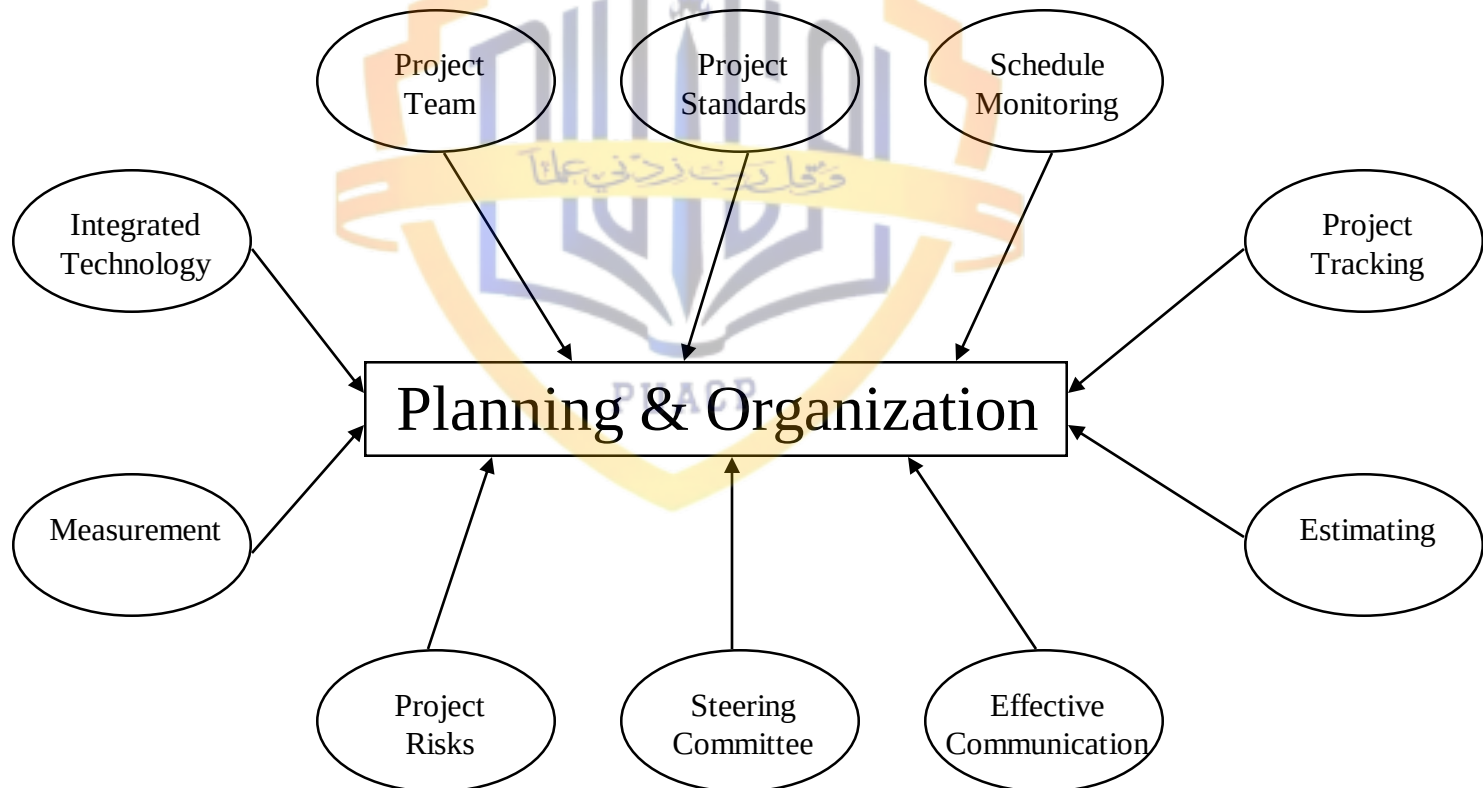
Process Assurance - 2

- Process assurance involves the interrelationships of several different components. Depending on how these are managed, they can have a major positive impact on the products
- Once an effective process assurance program is put in place and shown to be beneficial, then emphasis can be placed in making verification and validation strategies effective and in improving the quality of the products

Process Assurance - 3

- Successful process assurance is based on planning and organization
- There are several important aspects of planning and organization that must be considered before starting the project
- I'll show you a picture, which captures the components of planning and organization

Components of Planning and Organization



Project Team

- Project team is the project manager's only means of reaching the project goals
- Formation of project team is vital to success
- Size of the team depends on the size and complexity of the project
- Right mix of technical knowledge and experience
- Fostering of mutual respect within team and maintains good morals

Project Standards - 1

- Before the project is started, standards should be established for activities like requirements gathering, design development, and unit testing
- Standards should also be developed for quality control activities, like walkthroughs, reviews, and inspections

Project Standards - 2

- Many companies follow IEEE software engineering standards or they have their internally developed standards
- Standards should be flexible enough to be applied to large or small projects
- Any deviations from the standards should be approved by the project team and the reason for such deviation should be noted in the minutes of the project meetings

Schedule Monitoring - 1

- Stringent deadlines for the project are frequently established by management, end users, a project sponsor, or a client with no regard to the reality of achievement
- The project manager is then designated to meet unrealistic expectations of the project completion date
- For this reason, the project start date, milestones, and completion date should be negotiated upfront

Schedule Monitoring - 2

- If the unrealistic date is accepted and the project activities are then made to fit within this time frame, the quality of the project certainly will suffer
- The key to an “on-time” project lies in the ability to identify the critical path before starting the project

Schedule Monitoring - 3

- The critical path of a project is where problems that may affect the overall schedule are faced
- Develop systematic work breakdown structures which identify task groupings (tasks that can be combined together), task sequences, and entrance/exit criteria for each task

Schedule Monitoring - 4

- To define tasks, follow the guidelines of the system development methodology used by your organization
- In the absence of a development methodology, obtain copies of task lists and task dependencies from other projects and customize them to suit your needs of the current project

Schedule Monitoring - 5

- Clearly defined work breakdown structures will assist in selecting the correct skilled resources
- At the same time, using the breakdown structures also ensures that no activity is forgotten
- The technique of breaking down activities into smaller tasks takes an impossibly complex project and reorganizes it into manageable pieces under the direction of the project manager

Schedule Monitoring - 6

- Once you have defined the critical path, review the tasks and schedule with the project team members and other significantly impacted individuals

Schedule Monitoring - 7

- Since these people are the stakeholders and are affected by the project in one or more of the following ways:
 - Their budget is charged for all or part of the project
 - The department's resources are used by the project
 - The department has either existing projects or ongoing projects that are affected by the new project

Schedule Monitoring - 8

- Avoid the most common mistake of adding another resource to shorten or meet the schedule



Project Tracking

- Project tracking is an important activity in project management
- Project tracking tools should be used to monitor the progress of project schedule, and estimate resources

Estimation

- Realistic estimates allow you to discuss alternative approaches at the start of the project
- Estimates are not foolproof
- Allow time for resource management and unforeseen events, like the illness of a team member
- Revise estimates, and update plans

Effective Communication

- Effective communication between the management and project team is a critical interpersonal skill
 - Listening
 - Observing
 - Giving guidance
- Delegation of responsibility
- Informal negotiation

Steering Committee - 1

- A committee responsible for defining project policy, reviewing the project milestones, and evaluating risk factors must be established
- Members of the committee should represent all the impacted areas of the business. They should be knowledgeable enough to make informed technological decisions and be able to change the course if needed

Steering Committee - 2

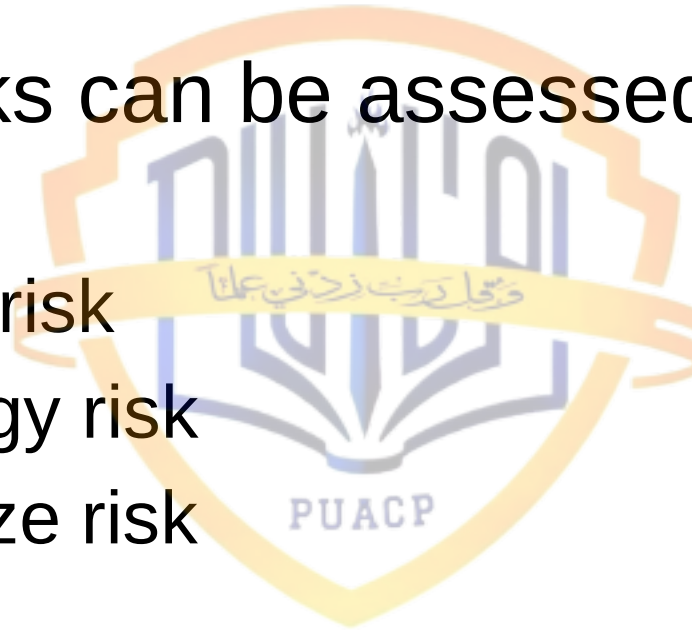
- It is responsible for
 - Estimating the time that will be required to maintain the system
 - Deciding on the type of support required from the operations for the running of the system
 - Deciding when the data will be available and how it will be managed, reported, and used
 - Forming a configuration control board (CCB) that manages the impact of changes

Project Risks - 1

- Every project has risks associated with them, some more than others
- There is need to identify and address the risk factors upfront
- All risk factors should be discussed with the project team, management, and users
- A risk mitigation policy needs to be developed

Project Risks - 2

- Project risks can be assessed in three categories
 - Business risk
 - Technology risk
 - Project size risk



Project Risks - 3

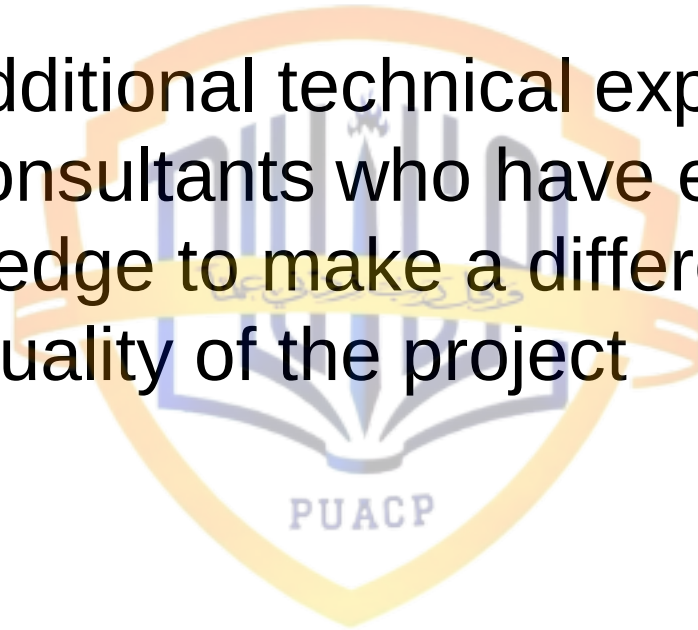
- Risks can be minimized by
 - Implementing controls from the initiation stage and by ensuring pre-established development standards are followed
 - Providing project management training
 - Reducing the scope of the project by incremental development or phased development

Project Risks - 4

- Technical risk is encountered when the project team utilizes a new technology like new hardware or new development methodology for the first time
- Technical risks can be controlled by
 - Appointing a qualified technical project leader
 - Implementing a strong, independent quality control group to evaluate the progress of the project and project deliverable

Project Risks - 5

- Getting additional technical expertise from outside consultants who have expertise and the knowledge to make a difference in the ultimate quality of the project



Measurement - 1

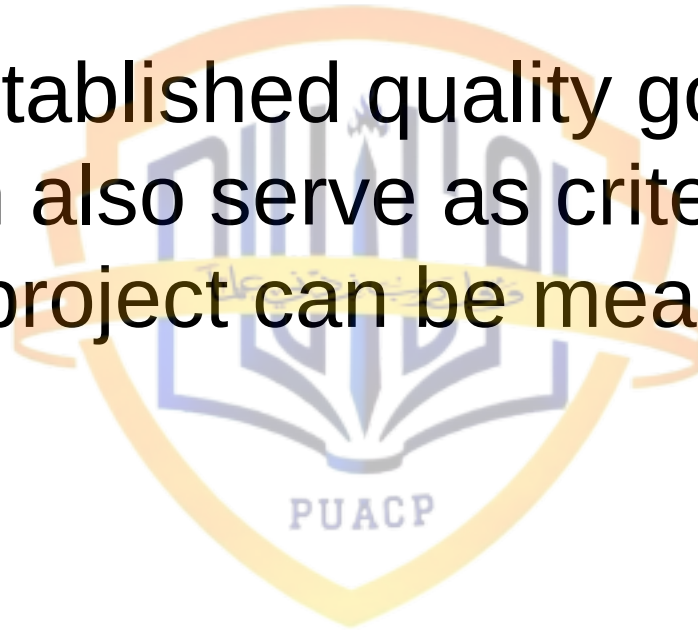
- Establishing measurement criteria, against which each phase of the project will be evaluated, is vital
- When exit criteria is well defined, it is sufficient to evaluate the outcome of each phase against the exit criteria and move forward

Measurement - 2

- If the outcome of each phase does not meet the performance criteria, the project manager should be able to control the project by evaluating the problems, identifying the deviations, and implementing new processes to address the deviations

Measurement - 3

- The pre-established quality goals for the project can also serve as criteria against which the project can be measured



Measurement - 4

- Processes should be established to
 - Enable the organization to address customer complaints
 - Give the organization statistics regarding the types of customer calls
 - Incorporate reporting and handling of customer problems
 - Enable management to make staffing decisions based on the number of customer calls

Integrated Technology - 1

- Strategy for Integrated Technology should be considered by management in relation to the other business needs
- This will empower the management to react to the operational needs of the business and, at the same time, take an inventory of the current status of various systems, projects, and the ability of technical staff to support any future projects

Integrated Technology - 2

- The IT trends, competitors, and demands of the customers should be visible to the management
- Parts of the new system that will be interfacing with existing system should be identified so that the impact can be evaluated
- If technology is new and not well understood, allowances to incorporate experiments should be made in the overall project plan and schedule

Causes of Failure in Process Assurance

- Lack of Management Support
- Lack of User Involvement
- Lack of Project Leadership
- Lack of Measures of Success



Symptoms of Process Failure - 1

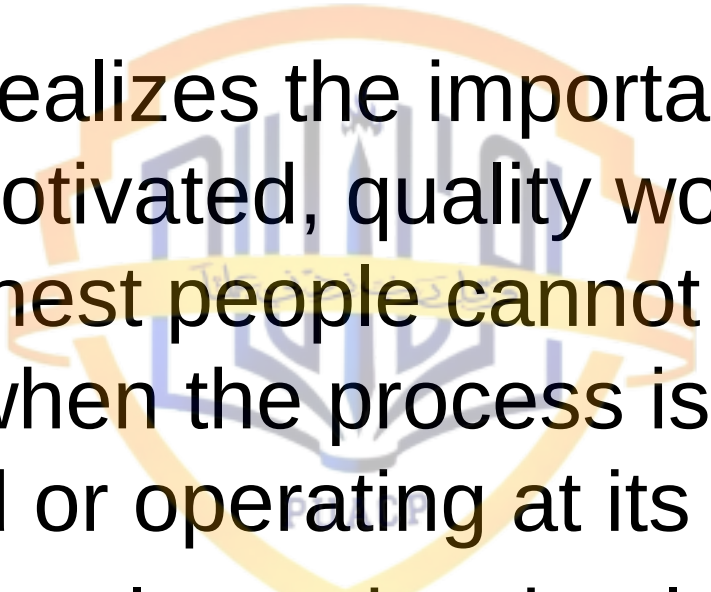
- Commitments consistently missed
 - Late delivery
 - Last minute crunches
 - Spiraling costs
- No management visibility into progress
 - You're always being surprised

Symptoms of Process Failure - 2

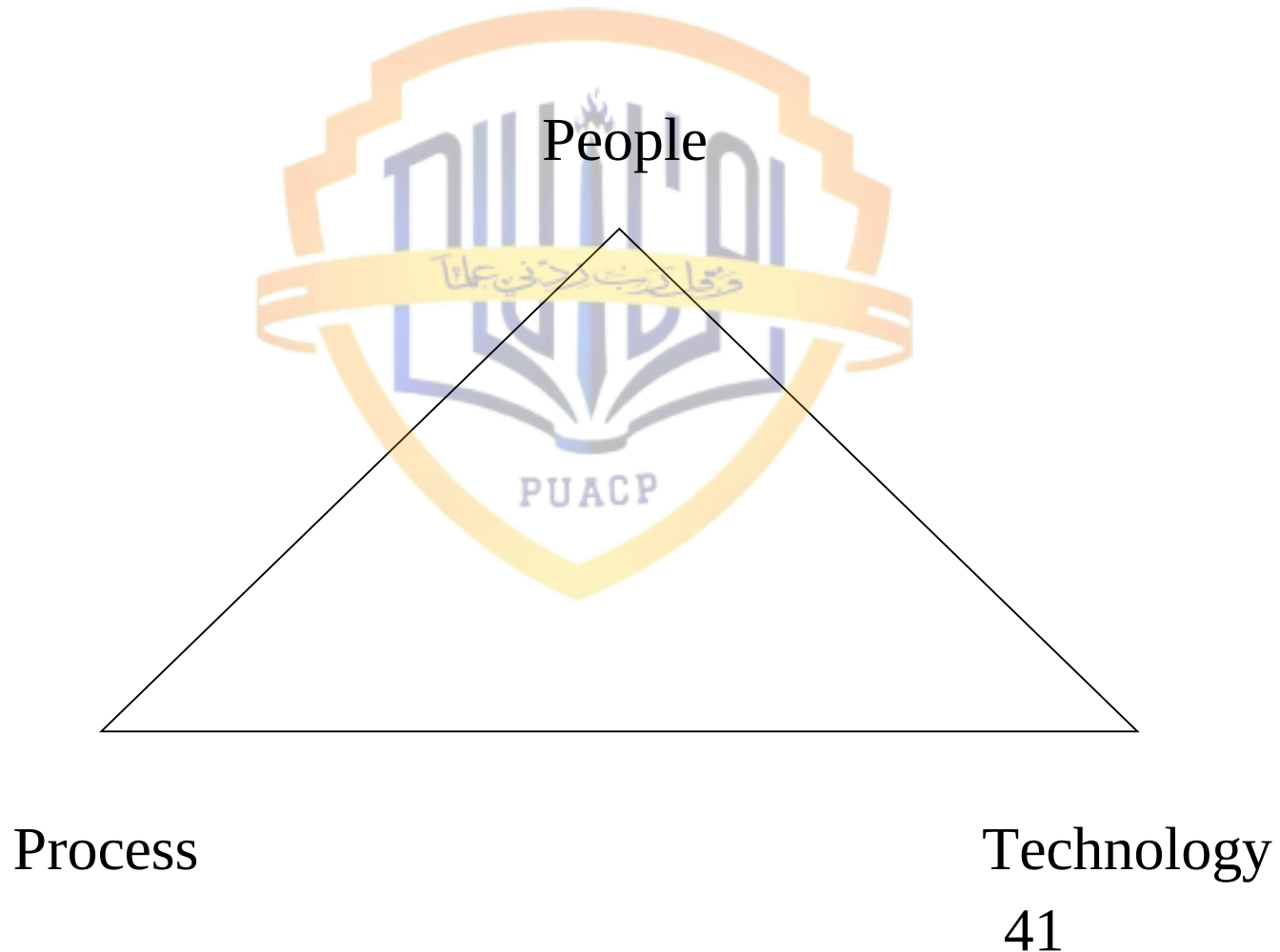
- Quality problems
 - Too much rework
 - Functions do not work correctly
 - Customer complaints after delivery
- Poor morale
 - People frustrated
 - Is anyone in charge?

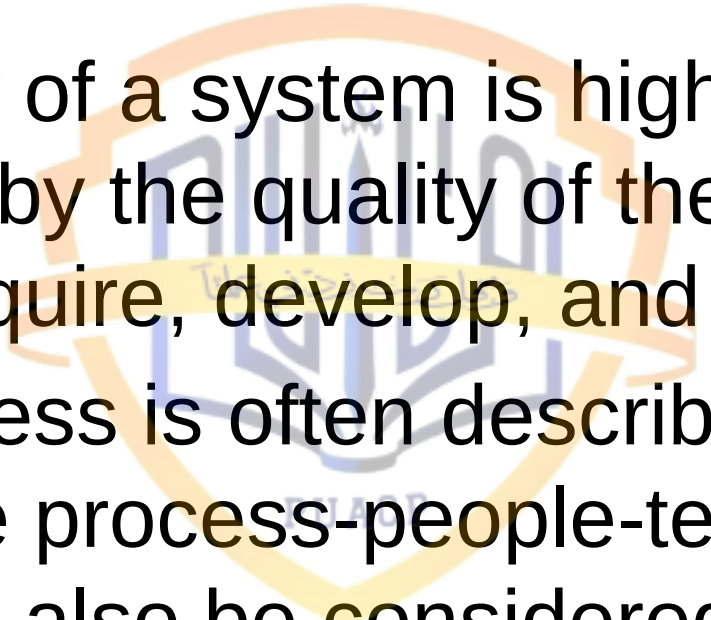
Common Misconceptions

- I don't need process, I have
 - Really good people
 - Advanced technology
 - An experience manager
- Process
 - Interfaces with creativity
 - Equals bureaucracy + regimentation
 - Isn't needed when building prototypes
 - Is only useful on large projects
 - Hinders agility in fast-moving markets
 - Costs too much

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- Everyone realizes the importance of having a motivated, quality work force but even our finest people cannot perform at their best when the process is not understood or operating at its best
 - Process, people, and technology are the major determinants of product cost, schedule, and cost

Process, People, and Technology



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- The quality of a system is highly influenced by the quality of the process used to acquire, develop, and maintain it
 - While process is often described as a node of the process-people-technology triad, it can also be considered the “glue” that ties the triad together

References

- Inroads to Software Quality by Alka Jarvis and Vern Crandall (Chapter 1)
- Software Quality: Analysis and Guidelines for Success by Capers Jones